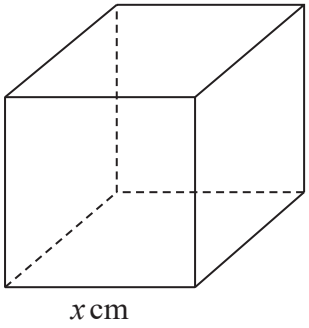
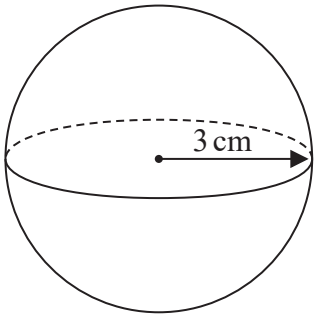
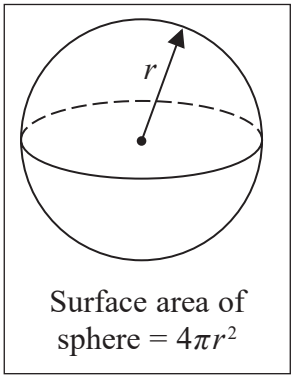
7 The diagram shows a cube with edges of length x cm and a sphere of radius 3 cm.







The surface area of the cube is equal to the surface area of the sphere.

Show that $x = \sqrt{k\pi}$ where k is an integer.

Cube: 6 faces, each area x^2, so area = $6x^2$

Sphere: $4\pi r^2 = 4\pi(3^2) = 36\pi$

Equal areas: $6x^2 = 36\pi$

Divide by 6: $x^2 = 6\pi$

$x > 0$, therefore $x = \sqrt{(6\pi)}$, so $k = 6$.

1 mark
cube area $6x^2$

1 mark
sphere area 36π

1 mark
equate the two areas

1 mark
$x = \sqrt{(6\pi)}$; $k = 6$

(Total for Question 7 is 4 marks)

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